

Designing and Building A Web Platform for Voice-Driven Community Libraries

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1. Introduction

College gave me critical texts that I never had time to fully explore. This platform will allow me and my community to collect what mattered, see how it connects, and preserve it beyond Yale.

Social media platforms optimize for viral reach and algorithmic engagement, minimizing the possibility of intimate knowledge-sharing within close communities. Influential apps like Instagram and TikTok strip content of personal context and make any act of communication or sharing a performance that can be tracked and compared through likes, views, and followers.

My project proposes an alternative. I seek to create a web platform that enables small, isolated communities to build collective knowledge archives through manual curation of media and text. In addition, every element is connected to an audio recording from the user, pushing oral traditions to the center of the experience. The system inverts contemporary social media design through three principles:

(1) focuses on distinct communities (groups exist independently and communities aren't connected), **(2) prioritizing oral traditions** (every item includes a personal narrative recording about why it matters), and **(3) human curation** (connections come from conversation, not algorithms).

This project draws on my background in HCI research and interface/product design to create an "anti-social media" platform. Rather than maximizing engagement, I will focus on carefully applying of friction and transparency to this community-first architecture. I have also deliberately constrained the scope; I will develop and deploy the app with my Yale community of peers as the sole community. Finally, the final site will serve as both a tool users can leverage and an installation piece for the art spring thesis exhibition.

2. Background / Related Works

Many of the most prevalent social media platforms weaponize "dark patterns" to maximize their profit and sneakily strip users of autonomy. Defined as "unethical design practices," dark patterns play a pivotal role in the user experience of today's Facebook, Instagram, TikTok, and X (formerly Twitter): infinite scroll, engagement-baiting notifications, obfuscated account deletion [1]. These patterns emerge from a business model that maximizes time-on-platform for ad revenue. Crucially, features like these prevent users from using the platforms with intention and moderation.

Intentional friction, understood to be purposeful "points of difficulty occurring during interaction," is a solution researchers believe can prevent app experiences from becoming addictive and mindless [2]. When implemented into a social network application that seeks to strengthen and reinforce

existing relationships and communities instead of fostering new ones, users can have an intimate and fulfilling app experience.

Network science shows that intimate support networks (approx five people) benefit from face-to-face interaction and correlate weakly with social media use, while larger networks (15-150 people) are shaped by digital platforms [3]. This suggests tools for intimate knowledge-sharing must sit in between these norms; they should leverage digital technology while emphasizing direct and meaningful interaction.

Existing tools don't support what this project needs. Goodreads focuses on individual reading habits; Are.na optimizes for discovery across users; Notion and Obsidian serve personal knowledge management. None enforce community boundaries or center testimony.

From a design perspective, a central principle of the project is that emphasizing the connections between separate texts and media helps illuminate every piece. It is this very framework that makes the Academy Award-nominated documentary *Soundtrack to a Coup d'Etat* so striking and successful—jazz songs from John Coltrane and Louis Armstrong play alongside advertisements from Tesla and Apple, as well as archival material from Patrice Lumumba's fight for Congolese independence. Sometimes the content overload feels jarring; other times, it feels smooth [4]. But regardless of how all the elements are presented together, the audience learns not only about “the bigger picture,” but also the nuances of each individual element. My project will seek to deliver a similar effect, bringing disparate pieces of social commentary and media together and fluctuating between smoothness and cacophony.

3. Problem Description

Design Challenge: How can I create a digital tool enabling small communities to build collective knowledge archives while preserving intimacy, personal testimony, and intentionality?

The platform succeeds if it enables a community to:

1. Build a shared library reflecting collective values and interests
2. Make visible the connections between works that shape the group's thinking
3. Preserve personal testimony alongside recommendations
4. Feel like a space for knowledge-sharing rather than performance

4. Planned Approaches

4.1 Data Model

Users: Email, name, bio

Items: Title, type, creator, link, audio_testimony, transcript, tags, added_by

Connections: Source_item, target_item, audio_description, created_by

Community: Name, email_whitelist

4.2 System Architecture

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Web Interface (Next.js + React)
|-- Add/Edit Items
|-- List View (search, filter)
|-- Graph Visualization (D3.js)
|-- Activity Feed
\-- Installation Mode (fullscreen)
    |
    v
Firebase Backend
|-- Firestore (NoSQL document database)
|   |-- users collection
|   |-- items collection
|   |-- connections collection
|   \-- community collection
|-- Firebase Auth (email whitelist)
\-- Firebase Storage (audio files)
    |
    v
Visualization Layer (D3.js force-directed graph)

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4.3 Core Features

1. **Item Management:** Add items with title, type, link, **audio testimony recording** (voice memo explaining “why this matters”), tags. Edit own items. Mark as private.
2. **Connection Creation:** Link any two items with audio or text description. View all connections to/from items.
3. **Library Views:**
 - List view: sortable table with search/filter
 - Graph view: interactive network (node size = connections, color = type/person)
4. **Activity Feed:** Recent additions and connections (last 20 actions)
5. **Installation Mode:** Fullscreen graph for exhibition, enhanced typography, auto-playing audio testimony as nodes are explored

4.4 Technical Stack

Backend: Firebase (Firestore NoSQL database + Firebase Auth + Firebase Storage)

Audio Transcription: OpenAI Whisper API (automatic transcription for searchability/accessibility)

Auth: Firebase Authentication with email whitelist authorization

Deployment: Vercel

4.5 Design Principles

Informed by HCI research on friction and anti-engagement patterns:

- **Audio-first testimony:** Voice recordings as primary testimony method, privileging oral tradition over text
- No engagement metrics (no likes, shares, view counts)
- No algorithmic sorting (chronological or user-controlled only)
- Visible authorship (every item/connection shows who added it)
- Mandatory testimony (can't add without audio explanation of why it matters)
- Community transparency (all activity visible to members)

Visual design draws on background in graphic design and typography, creating information-forward aesthetic that feels like archive or library catalog rather than social feed. Audio player design will emphasize listening as primary interaction.

4.6 Optional (Time Permitting)

- **AI-based transcript analysis:** Use GPT-4 or similar to analyze audio transcripts for automatic tag suggestions, theme extraction, and connection recommendations
- Enhanced filtering: multi-dimensional (type + person + tag)
- Export: JSON/PDF/SVG

5. Schedule / Timelines

Weeks 1-2 (Feb 3-14): Finalize data model, complete wireframes, advisor approval, dev environment setup

Milestone: Design approved, ready to build

Weeks 3-4 (Feb 17-28): Auth (email whitelist), item CRUD operations, basic list view with search/filter

Milestone: Can add, view, and manage items

Weeks 5-6 (Mar 3-14): Connection creation, graph visualization (D3.js), activity feed

Milestone: Full graph navigation functional

Week 7 (Mar 17-21): Installation mode (fullscreen view), initial typography/design polish

Milestone: Feature-complete MVP (since classes resume on the 23rd, having a MVP complete is crucial so that testing and data collection at large can begin)

Week 8 (Mar 24-28): First round user testing with 3-5 community members, bug fixes based on feedback

Milestone: Stable core system

Week 9 (Mar 31-Apr 4): Community content gathering through interviews, seed database with real data, observe usage patterns

Milestone: Platform populated with community knowledge

Week 10 (Apr 7-11): Design refinement based on real content, installation iterations, second round testing

Milestone: Installation ready for exhibition

Week 11 (Apr 14-18): Final writing, documentation, analysis of emergent patterns in knowledge graph

Milestone: Draft written doc complete

Week 12 (Apr 19-20): Final polish, submission preparation, advisor review

Deadline: All materials submitted April 20

Key Checkpoints: Feb 14 (wireframes), Feb 28 (core features), Mar 14 (full system), Mar 28 (first testing round), Apr 11 (installation ready), Apr 20 (submission)

6. Deliverables

1. **Functional web application** deployed for community access with all core features
2. **Source code repository** on GitHub with documentation and setup instructions
3. **Written documentation** covering technical implementation, design decisions grounded in HCI literature, analysis of emergent knowledge graph patterns, reflection on oral tradition in digital spaces
4. **Visual design assets** including wireframes, custom typography, system diagrams
5. **Final presentation** with live demo (3-5 minutes)

7. Concurrent Work

CPAR Art Component (Ongoing): Under the guidance of Art advisor Vamba Bility, I will complete the visual installation for final exhibit in coming months.

8. References

- [1] Mildner, T., et al. (2023). About Engaging and Governing Strategies: A Thematic Analysis of Dark Patterns in Social Networking Services. *CHI 2023*.
- [2] Cox, A. L., et al. (2016). Design Frictions for Mindful Interactions: The Case for Microboundaries. *CHI EA 2016*.
- [3] Hampton, K. N., et al. Core Networks, Social Isolation, and New Media. *Information, Communication & Society*.
- [4] Grimonprez, Johan, dir. Soundtrack to a Coup d'Etat. KINO LORBER, 2024.